Which of the above assumptions are valid?

(a) 1 and 3 (b) 2 and 4
(c) 2 and 3 (d) 1 and 4

Ans : (b) If you check assumption (3), it is not necessarily valid always, but only when a crisis (mutating crop disease as future climate change) happens. Hence, eliminate option (a) and (c). Assumption (1) talks about. "extinction of plant species" but the passage is about "food varieties". Assumption (2) is surely correct. So, (2) and (4) are right and answer is option (b).

LOGICAL & ANALYTICAL ABILITY

31. Consider the following Statements and Conclusions :

Statements

1. **Some rats are cats**
2. **Some rats are dogs**
3. **No dog is a cow.**

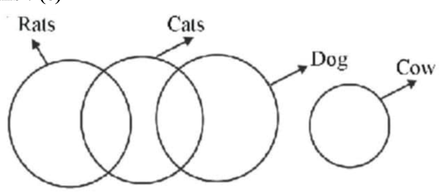
Conclusions:

I. **No cow is a cat**
II. **No dog is a rat**
III. **Some cats are rats.**

Which of the above conclusions is/ are drawn from the statements?

(a) I, II and III
(b) Only I and II
(c) Only III
(d) Only II and III

Ans : (c)

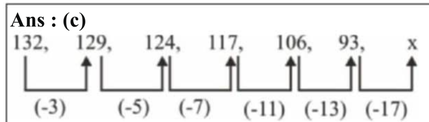


The conclusion "Some cats are rats" is drawn from the statements.

32. What is X in the sequence
132, 129, 124, 117, 106, 93, X

(a) 74 (b) 75
(c) 76 (d) 77

Ans : (c)



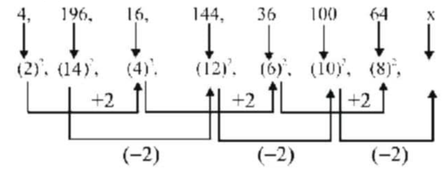
$\therefore x = 93 - 17$

$x = 76$

33. What is X in the sequence
4, 196, 16, 144, 36, 100, 64, X ?

(a) 48 (b) 64
(c) 125 (d) 256

Ans : (b)



$\therefore x = (8)^2$

$x = 64$

34. Consider the following sequence that follows some arrangement :
c-accaa-aa-bc-b
The letters that appear in the gaps are c-accaa-aa-bc-b

(a) abba (b) cbbb
(c) bbbb (d) cccc

Ans : (b) c **c** acc/aa **b** aa/ **b** bc **b** b
$\therefore$ The letters that appear in the gaps are cbbb.

35. Which year has the same calendar as that of 2009 ?

(a) 2018 (b) 2017
(c) 2016 (d) 2015

Ans : (d)

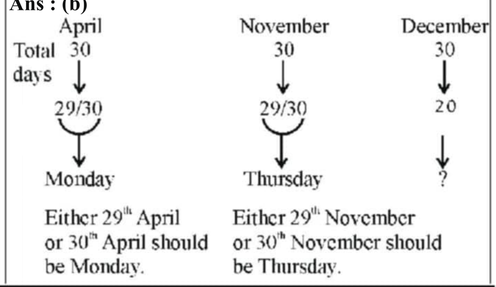
If the first day of 2009 is Thursday
The first day of 2010 is Friday (+1)
The first day of 2011 is Saturday (+1)
The first day of 2012 is Monday (+2)
2012 is a Leap year (+1)
The first day of 2013 is Tuesday
The first day of 2014 is Wednesday (+1)
The first day of 2015 is Thursday (+1)

Hence, the year of 2015 has the same calendar at that of 2009.

36. Mr 'X' has three children. The birthday of the first child falls on the 5th Monday of April, that of the second one falls on the 5th Thursday of November. On which day is the birthday of his third child, which falls on 20th December ?

(a) Monday (b) Thursday
(c) Saturday (d) Sunday

Ans : (b)



If monday falls on 29th April:-

Then, The remaining day of
(April + May + June + July + Aug + Sep + Oct + Nov)

1 + 31 + 30 + 31 + 31 + 30 + 31 + 30

$\Rightarrow \frac{215}{7} \Rightarrow 3\frac{5}{7}$ → Remainder

Friday falls on 30th November and Thursday falls 29th November.

Hence, Thursday is the birthday of his third child, which falls on 20th December.

April 29/30 → Monday
November 29/30 → Thursday, Friday
December 20 → Thursday

37. In the sequences 1,5,7,3,5,7,4,3,5,7, how many such 5s are there which are not immediately preceded by 3 but are immediately followed by 7?

(a) 1 (b) 2
(c) 3 (d) None of these

Ans : (a) Given sequences- 1,5,7,3,5,7,4,3,5,7
The number of required 5_s =1

38. If every alternative letter of the English alphabet from B onwards (including B) is written in lower case (small letters) and the remaining letters are capitalized, then how is the first month of the second half of the year written?

(a) JuLY (b) jULy
(c) jUly (d) jUlY

Ans : (d) According to question, the English alphabet-
b C d E f G h I j……Yz.
Hence, Small letter will be on even places and capital letters will be on odd places.
Then, The first month of the second half of the year is written as:- JULY

J U L Y
10 21 12 25

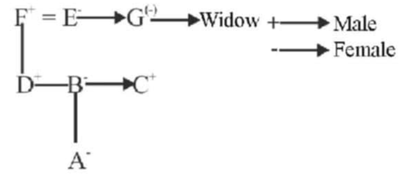
39. A joint family consists of seven members A, B, C,D,E,F and G with three females. G is a widow and sister-in-law of D's father F. B and D are siblings and A is daughter of B, C is cousin B. Who is E?

1. **Wife of F**
2. **Grandmother of A**
3. **Aunt of C**

Select the correct answer using the code given below-

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans : (d)



(i) E, wife of F → true
(ii) E, Grand mother of A → true
(iii) E, Aunt of C → true
$\therefore$ All three statements are true.

40. If $ means 'divided by'; @ means 'multiplied by; # means 'minus', then the value of 10#5@1$6 is,

(a) 0 (b) 1
(c) 2 (d) 9

Ans : (d) 10 # 5 @ 1 $ 5

$\Rightarrow 10 - 5 \times 1 \div 5$

$\Rightarrow 10 - 5 \times \frac{1}{5}$

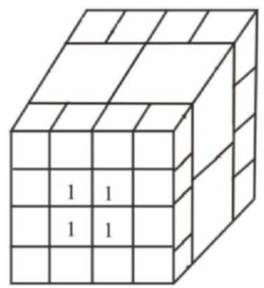
$= 10 - 1$

$= 9$

41. A solid cube is painted yellow, blue and black such that opposite faces are of same colour. The cube is then cut into 36 cubes of two different sizes such that 32 cubes are small and the other four cubes are big. None of the faces of the bigger cubes is painted blue. How many cubes have only on face painted ?

(a) 4 (b) 6
(c) 8 (d) 10

Ans : (c)



The number of cubes which have only on face printed is 8

Note: 4 from + side cube + 4 back side cube = 8 cube.

42. P, Q and R are three towns. The distance between P and Q is 60 km, whereas the distance between P and R is 80 Km. Q is in the West of P and R is in the South of P. What is the distance between Q and R ?